



TSK fuzzy system fusion at sensitivity-ensemble-level for imbalanced data classification

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ABSTRACT

Previous studies have shown that the performance of a classifier on imbalanced data heavily relies on informative objects lying in borderline or overlapping areas. In this study, we adopt objective sensitivity to detect the most informative objects based on 0-order Takagi-Sugeno-Kang Fuzzy System (0-TSK-FS), and accordingly propose a novel TSKfuzzy system fusion framework at interpretable sensitivity-ensemble-level (ISE-TSK-FS) to achieve both promising classification performance and reasonable interpretability on imbalanced datasets. Specifically, we assume that a small perturbation of an input object can be interpreted as its neighboring future testing objects. Therefore, object sensitivity w.r.t 0-TSK-FS can be evaluated by the expectation of squared output differences between the input object and the objects in its neighborhood. Being guided by such object sensitivity, we partition the majority class using clustering into different blocks and then construct an ensemble under-sampling process to select the most informative objects from each block iteratively. Moreover, to avoid over-fitting issues in assembling, a self-paced factor is introduced to constrain the number of low-sensitive objects but still keep their “skeleton” contribution. Extensive experiments on 7 synthetic datasets, 30 UCI datasets and one real medical case demonstrate ISE-TSK-FS’s promising performance and good interpretability on imbalanced data.

1. Introduction

The Takagi-Sugeno-Kang (TSK) fuzzy system has attracted much attention in the last few decades due to its excellent nonlinear approximation performance and well-known interpretability [1–4]. IF-THEN fuzzy rules, as basic components of TSK fuzzy systems, can map the fuzzy sets in the input space to a varying singleton, which are often considered simple prediction models. The TSK fuzzy system driven by these simple prediction models (fuzzy rules) can clarify the overall behavior of nonlinear dynamical systems, even they are very complex. Due to the nonlinear approximation ability, TSK fuzzy systems have been widely used in prognostication, industrial control and biomedical data analysis [5–8]. Recently, multiple studies have shown the advantages of ensemble learning [9, 10] in terms of performance and generalization ability. Some ensemble strategies such as stacking [11–13] have been successfully introduced into the modeling process of the TSK

model systems, making it expand from a shallow model to a deep or hierarchical structure. For example, Zhou *et al.* [12] proposed a deep TSK fuzzy system D-TSK-FS which contains several 0-order TSK fuzzy systems (0-TSK-FS) that are connected in a stacking way. The input of each component (except for the first component) consists of the original input features and the prediction results of previous layers as augmented features. The benefit arising from such stacked structure is that the augmented features can open the manifold structure of the original feature space, thereby improving D-TSK-FS’s generalization ability. Other similar ensemble learning studies on TSK fuzzy systems can be found in [11, 14].

Although D-TSK-FS and its variants in [11, 14] show excellent performance and generalization ability, they assume that the training dataset is balanced. However, imbalanced classification tasks are widespread in practice, especially in the biomedical field. Directly applying most machine learning classifiers including D-TSK-FS to handle

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the imbalanced dataset can result in a bias towards the majority class while ignoring the objects in the minority class, which are often the interest of study. Therefore, adapting the ensemble learning of TSK fuzzy systems to imbalanced classification tasks is vital. Various methods from the data level to the algorithm level have been proposed to solve the imbalanced classification problem, which will be briefly reviewed in Section II.A. Notably, many scholars believe that if the class-to-class margin is clear in imbalanced data, then the algorithm can still learn enough discriminative patterns from the data [13, 15]. However, if there is an overlapping distribution between classes, the classification performance can be significantly decreased. In addition to the overlapping factor, other data difficulties such as noise, outliers and rare data distribution could also challenge the imbalanced classification where different models show various sensitivity to them.

In this study, we propose a new *TSK fuzzy system fusion framework at interpretable sensitivity-ensemble-level (ISE-TSK-FS)* for imbalanced data analysis. Specifically, a new measurement of object sensitivity is introduced to measure the output fluctuation of an object, which can be viewed as an indicator of the output consistency between the input object and its nearby unseen objects. For instance, for a specific classifier, i.e., 0-TSK-FS in our case, if its predicted output of an object is inconsistent with the output of this object's nearby unseen neighbors, we consider that this object is difficult to classify by 0-TSK-FS correctly. That is to say, this object is highly sensitive to this classifier. Usually, objects in borderline or overlap are more likely to become high-sensitive ones, and the population of these objects reflects the level of class overlapping. Moreover, such objects could contain more informative patterns than low-sensitive objects which can contribute more to learning a promising classifier. Therefore, in the ensemble framework, object sensitivity distribution is beneficial to better guide the re-sampling process for imbalanced data classification. Accordingly, we propose an ensemble framework based on the object sensitivity distribution to iteratively under-sample the objects from the majority class in a wise way. At the same time, we introduce a self-paced factor to keep the “skeleton” contribution of low-sensitive objects so as to prevent overfitting problems.

Although ISE-TSK-FS seems to share the same structure as a recent method SPE [15], ISE-TSK-FS features its novelty in the sense of object sensitivity. Unlike SPE which uses the training error as object sensitivity, ISE-TSK-FS uses the input perturbation to generate future unseen testing objects in a neighborhood of the input object and calculates the output fluctuation between the input object and nearby unseen testing objects as object sensitivity. Such object sensitivity is more straightforward to compute. Compared with SPE, ISE-TSK-FS also becomes more compatible with testing objects. Moreover, our proposed method guarantees the model interpretability, thanks to its interpretable basic ensemble component based on the 0-TSK-FS fuzzy systems. The main contributions can be summarized from the following three perspectives.

- (1) A new measurement of object sensitivity compatible with testing objects is proposed, based on the assumption that a small perturbation of an input object can be interpreted as future testing objects in its neighborhood.
- (2) Under the supervision of the new object sensitivity, we take 0-TSK-FS as the basic component and construct an ensemble classifier to iteratively under-sample the most informative objects from the majority class so that both performance and interpretability can be guaranteed on imbalanced datasets.
- (3) A self-paced factor is introduced to reduce the population of low-sensitive objects. Therefore, only the “skeleton” contribution of low-sensitive objects is retained to prevent overfitting problems.

The rest sections are organized as: in Section 2, we briefly review some related work about imbalanced learning. In Section 3, we present the proposed sensitivity-ensemble-level-based TSK fuzzy system ISE-TSK-FS. In Section 4, we report and analyze the experimental results.

In Section 5, we conclude the study and indicate our future work.

2. Related work

In a binary classification task, class imbalance means that the objects in one class are significantly more than those in the other class. If we directly train a classical classifier on an imbalanced dataset, it may have a high probability of misclassifying objects in the minority class to the majority class. However, it is worth noting that, in real application scenarios, it is more valuable to correctly classify objects in the minority class. Take prognostics of radiotherapy as an example. Suppose we have a study cohort including 99 non-event patients and one patient having distant metastasis (event) after conformal radiotherapy. A classifier aims to minimize the training error on this cohort to meet its objective goal can produce 99% accuracy just by classifying all patients to the non-event class. With this classifier, although 99 patients having non-event are correctly predicted, the most crucial patient having event is misclassified, which might cause the radiotherapist to make an inappropriate radiotherapy plan. Therefore, in real application scenarios, how to train a promising classifier on imbalanced data to correctly classify the objects in the minority class is very meaningful and important. The existing imbalanced learning methods to overcome such an issue can be organized into three groups [16–18], the *data-level* group, the *algorithm-level* group and the *ensemble-level* group.

- 1) *Data-level* group: methods in this group usually employ over-sampling or under-sampling to balance the positive class and the negative class in the training dataset before model training. Over-sampling aims to produce some fictitious objects in accordance with the minority class's distribution to balance the majority class, while under-sampling seeks to eliminate some objects from the majority class to balance the minority class. SMOTE (Synthetic Minority Oversampling Technique) is a commonly used over-sampling method which produces fictitious objects by interpolating between existing objects in the minority class and their K -nearest minority neighbors [19]. For instance, Sutton *et al.* developed a SMOTE-assisted classifier to classify breast cancer PCR (Pathologic Complete Response) on MRI (Magnetic Resonance Imaging) post-neoadjuvant chemotherapy due to the imbalance between patients with PCR and without PCR [20]. Soon afterward, its variants like borderline-SMOTE [21], safe-level SOMTE [22], local neighborhood-based SMOTE [23] and deep SOMTE [24] were also proposed. For instance, to identify chronic active multiple sclerosis lesions, Zhang *et al.* developed imbalance-aware deep learning framework using a deep SMOTE strategy for automated identification of rim and lesions on QSM (Quantitative Susceptibility Mapping) [24]. NCRN (Neighborhood Cleaning Rule-based Under-sampling) is a well-known under-sampling method which eliminates objects from the majority class whose most neighbors are from the minority class [25]. In general, methods in the data-level group are distance-based, which may be effective when the class borderline is clear. However, when there is overlap between the majority class and the minority class, methods in the data-level group may aggravate / distort the overlap or omit some informative objects.
- 2) *Algorithm-level* group: methods in this group attempt to design penalty / regularized terms and integrate them into the objective to improve the ability of the classifier to perceive the minority class. For instance, based on AUC (area under the ROC curve), Wang *et al.* designed a penalty term towards the minority class and integrated it into the ELM (Extreme Learning Machine) classification framework to improve the perception ability towards the minority class [26]. Khan *et al.* designed a cost-sensitive loss function and embedded it into a deep neural network for robust feature learning of the minority class and the majority class [27]. The proposed method had good abilities of classification calibration and guess-aversion. In [28], Jiao

et al. employed cost-sensitive SVM to predict neonatal respiratory morbidity based on radiomics features extracted from imbalanced few-shot fetal lung ultrasound images. Compared with the data-level group, methods in the algorithm-level group have a significant merit. They do not alter the distribution of the training data, which lowers the computational cost of the training process. For all this, Wang *et al.* pointed out that, in the algorithm-level group, the assignment of the costs often determined by a cost matrix, which is often difficult to predefine or needs to be carefully tuned [29].

- 3) *Ensemble-level* group: methods in this group usually attempt to partition the imbalanced classification problem into several balanced classification problems, and then fuse these sub-problems in an ensemble manner. For example, in [30], SMOTE and adaptive boosting were combined to solve imbalance problems. In [31], the majority class was first randomly under-sampled in an unsupervised and supervised way, respectively, and then multiple Adaboost classifiers were trained on each balanced subset in an ensemble way. In [32], Tang *et al.* demonstrated the superiority of bagging-based ensemble and developed a voted ensemble framework for prognosis prediction of Head and Neck Squamous Cell Carcinoma. In our previous work [13], we proposed a deep-ensemble-level-based TSK fuzzy classifier based on stacked generalization principle to deal with the imbalance problems in dementia prediction. Generally, most ensemble-level methods follow a two-step training procedure: data balancing and ensemble learning. In data balancing, directly using under-sampling or oversampling may inherit the shortcomings we stated above. Also, in the ensemble learning procedure, how to deeply explore patterns from informative objects and stress noise objects to avoid overfitting is still a challenge.

As discussed, although much effort has been dedicated to imbalanced learning, the noted methods from the *data-level* group, the *algorithm-level* group and the *ensemble-level* group suffer from the following limitations and challenges.

- 1) Random sampling (under-sampling, over-sampling, etc.) destroys the pattern structure contained in training samples, especially ignoring the use of informative samples. In addition, random sampling may distort the sample distribution.
- 2) Although some data-level and algorithm-level methods use indicators like training errors to determine the informative objects, these indicators are not compatible with the testing objects.
- 3) Some ensemble-level methods also used TSK fuzzy systems as learning components. However, in the ensemble procedure, introducing intermediate output results into the feature space destroys the interpretability of fuzzy systems.

Motivated by the issues existing in previous studies, in this study, we aim to propose an interpretable sensitivity-ensemble-level-based TSK fuzzy system for imbalanced data analysis.

3. Proposed ISE-TSK-FS algorithm

In this section, first of all, some background concepts are given. Then the definition of object sensitivity is presented. Then based on this definition, the imbalance ensemble framework is constructed. Lastly, the detailed algorithm steps of ISE-TSK-FS are listed, and the time complexity is analysed.

3.1. Background concepts

0-TSK-FS is a commonly-used TSK fuzzy system due to its very simple

consequent. Therefore, in this study, it is also taken as the basic component of ISE-TSK-FS.

If a training set can be defined as $\chi = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where the input feature space $\mathbf{x}_i$ can be represented as $[x_{i1}, x_{i2}, \dots, x_{id}]^T \in R^d$ and $y_i \in R$ denote the output (label) space, then the fuzzy rule in 0-TSK-FS can be expressed as:

If x_{i1} is $\theta_1^k \wedge x_{i2}$ is $\theta_2^k \wedge \dots \wedge x_{id}$ is θ_d^k , then

$$f^k(\mathbf{x}_i) = \beta^k, k = 1, 2, \dots, K, \quad (1)$$

where θ_j^k represents a fuzzy set subscribed by the j -th feature for the k -th rule, and K presents the total number of used fuzzy rules. Each fuzzy rule aims to transform the fuzzy sets in the input feature space to a varying singleton denoted by $f^k(\mathbf{x}_i)$. After a series steps of operation and defuzzification, the output of $\mathbf{x}_i$, $y(\mathbf{x}_i)$ is computed by

$$y(\mathbf{x}_i) = \sum_{k=1}^K f^k(\mathbf{x}_i) \varphi^k(\mathbf{x}_i, \mu^k, \delta^k) = \sum_{k=1}^K \beta^k \varphi^k(\mathbf{x}_i, \mu^k, \delta^k), \quad (2)$$

in which the normalized fuzzy membership function $\varphi^k(\mathbf{x}_i, \mu^k, \delta^k)$ can be formulated as

$$\varphi^k(\mathbf{x}_i, \mu^k, \delta^k) = \frac{\tilde{\varphi}^k(\mathbf{x}_i, \mu^k, \delta^k)}{\sum_{k'=1}^K \tilde{\varphi}^{k'}(\mathbf{x}_i, \mu^{k'}, \delta^{k'})}. \quad (3)$$

In (3), $\tilde{\varphi}^k(\mathbf{x}_i, \mu^k, \delta^k)$ can be computed by $\prod_{j=1}^d \tilde{\varphi}_{\theta_j^k}^k(x_{ij})$. If the Gaussian

fuzzy membership function is adopted, then $\tilde{\varphi}^k(\mathbf{x}_i, \mu^k, \delta^k)$ can be further represented as

$$\tilde{\varphi}^k(\mathbf{x}_i, \mu^k, \delta^k) = \prod_{j=1}^d \exp\left(-\frac{(x_{ij} - \mu_j^k)^2}{2\delta_j^k}\right), \quad (4)$$

where $\mu^k = [\mu_1^k, \mu_2^k, \dots, \mu_d^k]$ and $\delta^k = [\delta_1^k, \delta_2^k, \dots, \delta_d^k]$ represent the kernel center vector and the kernel width vector, respectively.

From (2), we see that 0-TSK-FS is actually equivalent to a single layer feedforward neural network, as shown in Fig. 1, where the normalized fuzzy membership function $\varphi^k(\mathbf{x}_i, \mu^k, \delta^k)$ can be taken as the active function and β^k can be taken as the output weight of each hidden node.

According to [33], when μ and δ in the active function are determined by clustering, e.g., FCM, the optimization of 0-TSK-FS is actually a ridge regression problem, as shown in (5), which can be quickly solved by LLM (least learning machine) [34],

$$\min_{\beta} \left(\frac{1}{2} \beta^T \beta + C \sum_{i=1}^N \xi_i \right) \quad (5)$$

$$s.t. (\varphi^1(\mathbf{x}_i, \mu^1, \delta^1), \varphi^2(\mathbf{x}_i, \mu^2, \delta^2), \dots, \varphi^K(\mathbf{x}_i, \mu^K, \delta^K)) \beta^T = y_i + \xi_i.$$

where C is a user-defined constant. The analytical solution to (5) is computed by

$$\beta^T = \Omega^T \left(\Omega \Omega^T + \frac{1}{2C} \mathbf{I} \right)^{-1} \mathbf{y}, \quad (6)$$

where $\Omega = [\Omega_1, \Omega_2, \dots, \Omega_N]^T$, $\Omega_i = [\varphi^1(\mathbf{x}_i, \mu^1, \delta^1), \varphi^2(\mathbf{x}_i, \mu^2, \delta^2), \dots, \varphi^K(\mathbf{x}_i, \mu^K, \delta^K)]$, $\mathbf{y} = [y_1, y_2, \dots, y_N]^T$, and $\mathbf{I}$ is a N by N identity matrix. The training of 0-TSK-FS via LLM is described in Algorithm 1.

Compared with BP [35, 36] which needs to learn parameters both in the output and hidden layers, we find that LLM only need to learn β in the output layer. Since 0-TSK-FS is equivalent to a single layer

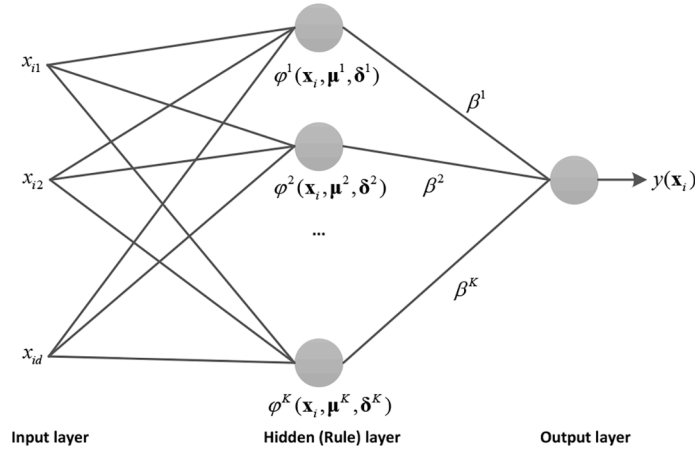


Fig. 1. Single layer feedforward neural network.

feedforward neural network and the parameters in its antecedent can be determined by clustering, we introduce LLM to ISE-TSK-FS for the quick training of O-TSK-FS, which is served as the basic component of ISE-TSK-FS.

3.2. A new measurement of object sensitivity

Originally, *object sensitivity* is defined as a measure of the probability that the object can be correctly classified by the classifier. The smaller the sensitivity, the easier it is for the classifier to determine the class. In [15], object sensitivity is specified by the training error w.r.t. a specific trained classifier. The absolute error function, the squared error function and the cross-entropy error function are usually used to compute the training error. Although this sensitivity specification method sounds reasonable, it only considers how to explore pattern information from

of our new definition is based on this assumption that the input object and its nearby unseen objects should have the same output. Compared with directly using the training error to specify object sensitivity, the new measurement can tentatively explore sensitivity contributions from testing objects.

For a given input object $\mathbf{x}_b$, its ε -neighborhood $\Omega_\varepsilon(\mathbf{x}_b)$ which covers all possible unseen testing objects can be defined as follows,

$$\Omega_\varepsilon(\mathbf{x}_b) = \{\mathbf{x} | \mathbf{x} = \mathbf{x}_b + \mathbf{z}, |\mathbf{z}_j| \leq \varepsilon, j = 1, 2, \dots, d\}, \quad (7)$$

where $\mathbf{z} = [z_1, \dots, z_j, \dots, z_d]$ is a perturbation vector also represented in the input feature space, ε denotes the upper bound to the maximum expected difference between $\mathbf{x}_b$ and possible unseen testing objects under investigation. Thus, the sensitivity of $\mathbf{x}_b$ can be defined as the expected errors between all object in $\Omega_\varepsilon(\mathbf{x}_b)$ and $\mathbf{x}_b$, that is,

$$\begin{aligned} \text{Sens}(\mathbf{x}_b) &= \int_{\Omega_\varepsilon(\mathbf{x}_b)} (f(\mathbf{x}_b) - f(\mathbf{x}))^2 p(\mathbf{x}) d\mathbf{x} = \int_{\Omega_\varepsilon(\mathbf{x}_b)} (f(\mathbf{x}_b)^2 - 2f(\mathbf{x}_b)f(\mathbf{x}) + f(\mathbf{x})^2) p(\mathbf{x}) d\mathbf{x} \\ &= f(\mathbf{x}_b)^2 - 2f(\mathbf{x}_b) \int_{\Omega_\varepsilon(\mathbf{x}_b)} f(\mathbf{x}) p(\mathbf{x}) d\mathbf{x} + \int_{\Omega_\varepsilon(\mathbf{x}_b)} f(\mathbf{x})^2 p(\mathbf{x}) d\mathbf{x} = f(\mathbf{x}_b)^2 - 2f(\mathbf{x}_b)\Theta_1 + \Theta_2. \end{aligned} \quad (8)$$

informative objects in the training set.

In the context of supervised pattern recognition problems, we have common knowledge that a perturbation of an input object can be interpreted as unseen objects nearby the input object, so that for a smaller perturbation, those unseen objects in the future may be recognized like training objects. Based on such knowledge, we redefine object sensitivity as the output fluctuation between an input object and its nearby unseen objects obtained by a slight perturbation. The rationality

We assume that all input features are independent to each other. Thus, with a trained O-TSK-FS, Θ_1 and Θ_2 in (8) can be defined as follows,

$$\begin{aligned} \Theta_1 &= \int_{\Omega_\varepsilon(\mathbf{x}_b)} f(\mathbf{x}) p(\mathbf{x}) d\mathbf{x} = \frac{1}{(2\varepsilon)^d} \int_{\Omega_\varepsilon(\mathbf{x}_b)} f(\mathbf{x}_b + \mathbf{z}) d\mathbf{x} = \frac{1}{(2\varepsilon)^d} \int_{\Omega_\varepsilon(\mathbf{x}_b)} \sum_{k=1}^K \beta_k \exp\left(\sum_{i=1}^d (z_i + x_{bi} - \mu_{ki})^2 / -2\sigma_k^2\right) d\mathbf{x} \\ &= \left(\frac{\sqrt{\pi}}{(2\sqrt{2}\varepsilon)^d}\right)^d \sum_{k=1}^K \beta_k \prod_{i=1}^d \left(\sigma_k \left(\text{erf}\left(\frac{x_{bi} - \mu_{ki} + \varepsilon}{\sqrt{2}\sigma_k}\right) - \text{erf}\left(\frac{x_{bi} - \mu_{ki} - \varepsilon}{\sqrt{2}\sigma_k}\right)\right)\right), \end{aligned} \quad (9)$$

$$\begin{aligned}
\Theta_2 &= \int_{\Omega_e(\mathbf{x}_b)} f(\mathbf{x})^2 p(\mathbf{x}) d\mathbf{x} = \frac{1}{(2\varepsilon)^d} \int_{\Omega_e(\mathbf{x}_b)} f(\mathbf{x}_b + \mathbf{z})^2 d\mathbf{x} = \frac{1}{(2\varepsilon)^d} \int_{\Omega_e(\mathbf{x}_b)} \sum_{k=1}^K \beta_k \exp \left(\sum_{i=1}^d (z_i + x_{bi} - \mu_{ki})^2 / -2\sigma_k^2 \right) d\mathbf{x} \\
&= \left(\frac{\sqrt{\pi}}{4\varepsilon} \right)^d \sum_{k,g=1}^K \left(\left(\sqrt{2\sigma_k^2 \sigma_g^2 (\sigma_k^2 + \sigma_g^2)} / (\sigma_k^2 + \sigma_g^2) \right)^d \prod_{i=1}^d \left(\begin{aligned} &\text{erf} \left(\frac{(\sigma_k^2 + \sigma_g^2)(x_{bi} + \varepsilon) - (\sigma_k^2 \mu_{ki} + \sigma_g^2 \mu_{gi})}{\sqrt{2\sigma_k^2 \sigma_g^2 (\sigma_k^2 + \sigma_g^2)}} \right) \\ & - \text{erf} \left(\frac{(\sigma_k^2 + \sigma_g^2)(x_{bi} - \varepsilon) - (\sigma_k^2 \mu_{ki} + \sigma_g^2 \mu_{gi})}{\sqrt{2\sigma_k^2 \sigma_g^2 (\sigma_k^2 + \sigma_g^2)}} \right) \end{aligned} \right) \right)
\end{aligned} \quad (10)$$

Additionally, we have no prior knowledge about unseen testing objects in $\Omega_e(\mathbf{x}_b)$, thus, we also assume that all unseen objects have the same probability to appear in future. That is, $p(\mathbf{x})$ in (8) can be calculated by $p(\mathbf{x}) = \frac{1}{(2\varepsilon)^d}$.

According to the sensitivity w.r.t. a specific classifier, it is very easy to imagine that all objects in a training set can be divided into three categories, *trivial* objects, *noise* objects and *boundary* objects.

Trivial objects refer to those samples that are easily determined by the classifier. That is, each trivial object has very low sensitivity and hence contributes a tiny sensitivity to the classifier. However, the overall contribution of trivial objects cannot be neglected due to the large population. In our ensemble learning process, it is expected that a small proportion of trivial objects should be kept in order to represent the “skeleton” of their distribution such that overfitting problems can be prevented.

Noise objects refer to the indistinguishable overlapping samples and outliers which are very sensitive to the classifier. That is, each noise object has a very high sensitivity and hence the total sensitivity contribution is very huge. In our ensemble learning process, we should stress these objects because enforcing to learn such objects could lead to serious overfitting problems.

Except for trivial objects and noise objects, the rest are objects.

Boundary objects are informative for a classifier to learn the decision boundary. Therefore, in our ensemble learning process, we should strengthen the weight of such boundary objects.

3.3. Sensitivity-ensemble-level framework

To well explore informative objects in the training set based on the proposed measurement of object sensitivity, we design a three-component imbalanced ensemble framework, as shown in Fig. 2.

Component 1: This component is used to train an initial 0-TSK-FS via random under-sampling (RUS).

Component 2: This component is used to partition the majority class into different block and compute the average sensitivity of each block. To be specific, we find the ε -neighborhood set of each training object in the majority class and compute the corresponding object sensitivity based on (8), (9) and (10). Then all training objects are partitioned into L blocks ($B_1, B_2, \dots, B_L$) by the K -means clustering method according to their object sensitivity values. Finally, we compute the average object sensitivity of each block by $SL_l = \sum_{s \in B_l} \text{Sen}(\mathbf{x}_s) / |B_l|$. In this way, each block indicates a specific sensitive level.

Component 3: This component is used to explore patterns from informative objects. To this end, we introduce the concept of “sensitivity reconciliation” and a self-paced training procedure.

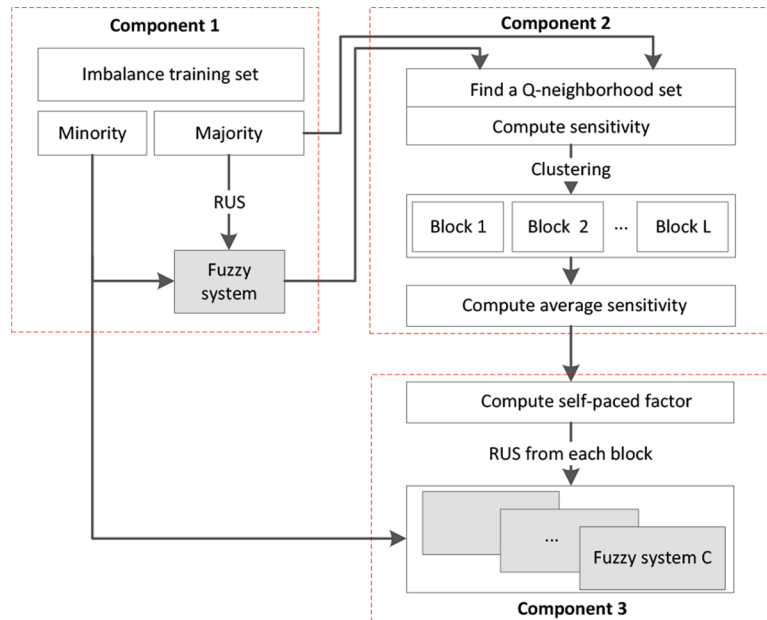


Fig. 2. Framework of ISE-TSK-FS.

- (1) *Sensitivity reconciliation*: the concept of “reconciliation” firstly comes from batch learning of neural networks. For example, in [37], the authors reconciled the gradient contribution of each batch in gradient-based optimization. Inspired by the idea of “reconciliation”, in this component, based on the block partition in component 2, we under-sample $\frac{1/SL_i}{\sum_{m=1}^L (1/SL_m)} |U_p|$ objects from each block to reconcile the sensitivity contribution of each block. That is to say, sensitivity reconciliation can keep the sensitivity contribution of each block consistent.
- (2) *Self-paced factor*: with 0-TSK-FS gradually fitting the training objects, the number of trivial (less informative) objects will increase significantly. That is to say, sensitivity reconciliation without supervision will select excessive trivial objects during under-sampling, thereby leading to overfitting. To address this issue, we introduce a self-paced factor α to supervise the sensitivity reconciliation-based under-sampling. That is to say, after the majority class is partitioned into different block, we under-sample $\frac{1/(SL_i+\alpha)}{\sum_{m=1}^L (1/(SL_m+\alpha))} |U_p|$ objects from each block to reconcile the sensitivity contribution of each block, where the self-paced factor α is defined as $\tan(i\pi/2C)$, i represents the i th ensemble procedure and C represents the total number of ensemble. As the self-paced factor goes large, our method ISE-TSK-FS pays more attention to those objects which have larger sensitivity values. Moreover, another merit of the self-paced factor is that even it goes very large, a reasonable size of trivial objects can still be kept as the “skeleton” so that the overfitting can be avoided.

Finally, we ensemble all learned 0-TSK-FS in a weighting way. That is, we record the training error as the weight of each classifier. According to Fig. 2, we list the detailed steps of ISE-TSK-FS, as shown in Algorithm 2.

Although the framework of ISE-TSK-FS is similar to SPE, they are totally different. SPE works well in scenarios of massive data. But in scenarios of small data, due to the adopted equal interval division strategy, SPE may generate some “empty blocks”. Therefore, under-sampling from all blocks may not balance the minority class. Therefore, in this study, clustering is employed to partition the majority class into different blocks according to the object sensitivity, which can well address the issue of “empty blocks”.

3.4. Learning algorithm of ISE-TSK-FS

Detailed algorithm steps are listed in Algorithm 2.

Remark: In step 6, the common random cluster center initialization strategy is replaced by a non-random strategy as given in Algorithm 3 in order to improve the robustness of ISE-TSK-FS. In addition, in antecedent learning of 0-TSK-FS, the fixed initialization strategy is also used to serve the same purpose. More information of robustness analysis can be found in Section 4.4.

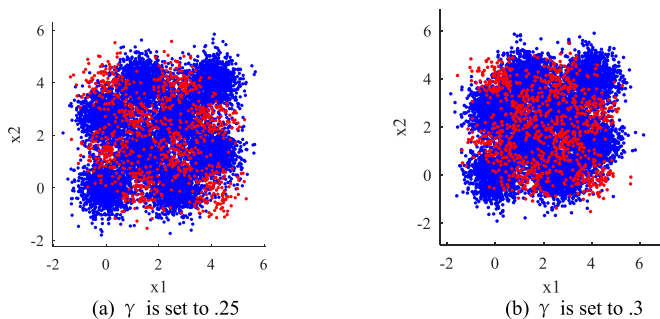


Fig. 3. Imbalanced synthetic dataset.

3.5. On training time complexity

Let us first observe the time complexity of 0-TSK-FS training. From Algorithm 1, we see that the training procedure of 0-TSK-FS contains the FCM clustering process and the computation of (6). Thus, the upper bound of the asymptotic complexity is $O(N^3)$. Hence, when the number of training objects is very large, the computation of (6) will consume a lot. To reduce the time complexity of 0-TSK-FS, by introducing LLM (Least Learning Machine) [34], (6) can be replaced by

$$\beta^T = \left(\Omega^T \Omega + \frac{1}{2C} \mathbf{I} \right)^{-1} \Omega \mathbf{y}, \quad (11)$$

which only needs $O(K^3)$ time complexity, where K denotes the number of fuzzy rules. Usually, K is far less than N . During each ensemble procedure, the time complexity of K -means is $O(N_{U_n} L d)$, where N_{U_n} is the size of the majority class and d denotes the number of features each object contains. Therefore, the final upper bound of the asymptotic complexity of ISE-TSK-FS can be computed by $O(K^3 + C N_{U_n} L d)$.

4. Experimental studies

We generate 7 imbalance synthetic datasets and introduce 30 real-world datasets with different imbalance ratios to evaluate our method. Additionally, several benchmarking methods are introduced for comparison studies. Finally, we statistically report the experimental results and analyze parameter sensitivity and interpretability.

4.1. Datasets

We generate an imbalanced synthetic dataset containing 16 Gaussian components. All components share the same covariance matrix $\gamma \times [1 \ 0; 0 \ 1]$, where γ can control the degree of the overlaps between the majority class and the minority class. The size of minority class is 1000 and the majority is 100. In our experiments, we generate 7 synthetic datasets with different γ from [0.05 0.07 0.10 0.15 0.20 0.25 0.30]. Fig. 3 illustrates two datasets with $\gamma = 0.25$ and $\gamma = 0.3$, respectively.

Additionally, we also introduce 30 real-world datasets with different imbalance ratios from the KEEL imbalance repository (<https://sci2s.ugr.es/keel/imbalanced.php>), as shown in Table 1, where IR denotes the imbalance ratio.

4.2. Experimental settings

For fair studies, we introduce several imbalance strategies/frameworks including RUS [38], CUS [39], SMOTE [21], ABSERR [15] and SPE [15]. The classifier for all strategies/frameworks is 0-TSK-FS. 0-TSK-FS is also used as the baseline. In addition, another ensemble-level-based TSK fuzzy system IDE-TSK-FC [13] is introduced for comparison studies.

BASELINE: 0-TSK-FS is adopted as the baseline in our experimental studies, where K is determined by cross-validation from [2, 4, 6, ..., 50].

RUS: RUS is the random under-sampling framework. It randomly selects samples from the majority class to balance the minority class.

CUS: CUS is the cluster-based under-sampling framework. It uses medoid-based clustering techniques to cluster the majority class into many small clusters and then selects medoids of clusters to balance the minority class.

SMOTE: SMOTE is an over-sampling framework. It produces fake objects in the minority class to balance the minority class.

ABSERR: ABSERR is an ensemble-level method which takes the absolute training error as object sensitivity.

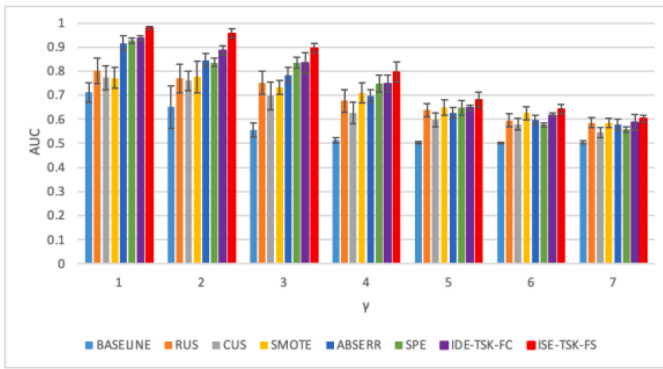
SPE [15]: SPE is an ensemble-level method which takes the absolute training error as object sensitivity.

IDE-TSK-FC [13]: IDE-TSK-FC is a deep-ensemble-level-based TSK fuzzy classifier based on the stacked generalization principle.

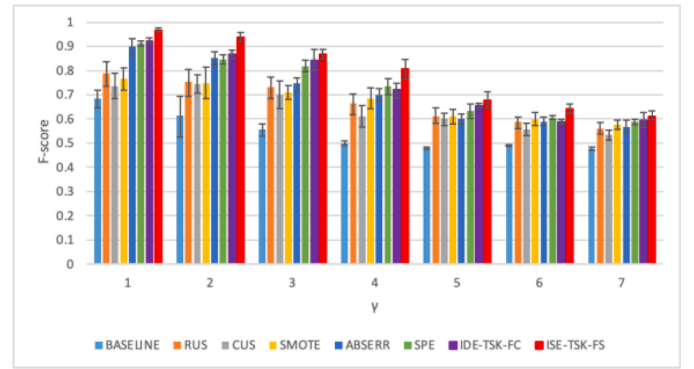
Table 1

Real-world imbalanced datasets.

Datasets	#Samples/ #Features	IR	Datasets	#Samples/ #Features	IR	Datasets	#Samples/ #Features	IR
Pima (Data-01)	768/8	1.87	Vehicle3 (Data-02)	846/18	2.99	Haberman (Data-03)	306/3	2.78
Ecoli1 (Data-04)	336/7	3.36	Yeast3 (Data-05)	1484/8	8.1	Ecoli3 (Data-06)	336/7	8.6
Glass016vs2 (Data-07)	192/9	1.29	Yeast1vs7 (Data-08)	459/7	14.3	Abalone918 (Data-09)	731/8	16.4
Yeast2vs8 (Data-10)	482/8	23.1	Yeast4 (Data-11)	1484/8	28.1	Yeast1289vs7 (Data-12)	947/8	3.57
Yeast6 (Data-13)	1484/8	41.4	Abalone19 (Data-14)	4174/8	129.44	Flare-F (Data-15)	1066/11	23.79
Krvskzeroonevsdraw (Data-16)	2901/6	26.63	Krvskthreevsseven (Data-17)	2935/6	35.23	Winequalityred8vs6 (Data-18)	656/11	35.44
Winequalitywhite3vs7 (Data-19)	900/11	44.0	Winequalityred8vs67 (Data-20)	855/11	46.5	Abalone19vs10111213 (Data-21)	1622/8	49.69
Krvskzerovseight (Data-22)	1460/6	53.07	Winequalitywhite39vs5 (Data-23)	1482/11	58.28	Poker89vs6 (Data-24)	1485/10	58.4
Shuttle2vs5 (Data-25)	3316/9	66.67	Abalone20vs8910 (Data-26)	1916/8	72.69	Kddcupbufferoverflowvsback (Data-27)	2233/41	75.67
Krvskzerovsfifteen (Data-28)	2193/6	8.22	Poker89vs5 (Data-29)	2075/10	82.0	Poker8vs6 (Data-30)	1477/10	85.88



(a) Classification performance in terms of AUC



(b) Classification performance in terms of F-score

Fig. 4. Classification performance on 7 synthetic datasets.

AUC and F-score [26] are employed as the indicators to evaluate the classification performance of each method. On each dataset, each method is executed for 30 tries, the average AUC and F-score and the corresponding standard deviations are recorded.

The experiments are conducted on a PC with Inter® Core (TM) i7-9700 @ dual 3.00 GHz and 32 G RAM. The coding platform is MATLAB R2016b.

4.3. Performance evaluation

The classification performance in terms of AUC and F-score of ISE-TSK-FS and the comparing algorithms on 7 synthetic datasets is shown in Fig. 4. From Fig. 4, we see that ISE-TSK-FS achieves the best performance on all synthetic datasets. It also can be found that *Ensemble-level* based algorithms perform better than *data-level* based algorithms.

The results on 30 real-world datasets are also given in terms of AUC and F-score, as shown in Tables 2 and 3, where the best performance is highlighted in bold. Also, to clearly observe the differences between ISE-TSK-FS and the benchmarking methods, we plot AUC (also F-score) against different datasets in Fig. 5. We observe that, in terms of AUC, ISE-TSK-FS achieves the best classification performance except on the datasets Data-12, Data-15, Data-23, Data-24, Data-26 and Data-27. In terms of F-score, ISE-TSK-FS wins the best performance except on the datasets Data-04, Data-21, Data-24, Data-26 and Data-27. These findings can also be clearly observed from Fig. 5. In addition, we record the running time of ISE-TSK-FS w/o LLM in terms of CPU seconds, as shown in Fig. 6. It can be found that LLM reduces the time complexity of ISE-TSK-FS significantly.

Experimental results in terms of comparative analyses indicate us

that ISE-TSK-FS performs better upon most occasions on the 7 synthetic datasets (Fig. 4) and 30 UCI datasets (Fig. 5, Tables 2 and 3). What is more, consistent findings can be found from statistical analyses (Tables 4 and 5). The lower CPU seconds also indicates the power of the effective optimization method LLM. All these findings demonstrate how great ISE-TSK-FS is. Notably, it is significant to remember that the current evidence depends on the following specifics:

- (1) Different from the original concept of object sensitivity which measures the amount of pattern information objects therein [15], a new measurement of object sensitivity is designed in this study based on the common knowledge that a small perturbation of an input object can be interpreted as future testing objects in its neighborhood. That is, our object sensitivity is compatible with testing objects. Our superiority can be demonstrated by comparing the classification results (Tables 4 and 5) obtained by ABSERR and ISK-TSK-FS, where ABSERR uses the old object sensitivity defined in [15].
- (2) Based on the new object sensitivity, our imbalanced ensemble framework enables effective partitioning of the majority class into multiple reasonably-sized blocks. Through a concordant & self-paced sampling technique, it selectively operates on objects with greater sensitivity values (i.e., higher misclassification rates) and offers better learning capability from informative borderline objects while retaining the general "skeleton" of object distribution.
- (3) Although the framework of ISE-TSK-FS is similar to SPE, they are totally different. SPE works well in scenarios of massive data. But

Table 2

Classification performance on 30 real-life datasets in terms of AUC and the standard deviation.

Datasets	0-TSK-FS	RUS	CUS	SMOTE	ABSERR	SPE	IDE-TSK-FC	ISE-TSK-FS
Data-01	0.6866±0.0312	0.7258±0.0122	0.7033±0.0270	0.7264±0.0308	0.7330±0.0148	0.7313±0.0128	0.7231±0.0236	0.7434±0.0240
Data-02	0.5616±0.0191	0.6774±0.0388	0.6551±0.0167	0.6831±0.0286	0.6851±0.0279	0.7013±0.0239	0.6987±0.0211	0.7054±0.0316
Data-03	0.5760±0.0390	0.6091±0.0430	0.5511±0.0544	0.6089±0.0400	0.6367±0.0496	0.6224±0.0326	0.6198±0.0432	0.6415±0.0424
Data-04	0.8049±0.0364	0.8756±0.0156	0.8726±0.0228	0.8810±0.0226	0.8792±0.0261	0.8761±0.0141	0.8878±0.0187	0.8950±0.0181
Data-05	0.7914±0.0298	0.9100±0.0126	0.8624±0.0183	0.9163±0.0118	0.9069±0.0300	0.9029±0.0210	0.8976±0.0254	0.9194±0.0159
Data-06	0.7528±0.0493	0.8561±0.0646	0.8724±0.0419	0.8782±0.0346	0.8170±0.0726	0.8782±0.0456	0.8548±0.0566	0.8885±0.0507
Data-07	0.4965±0.0055	0.6056±0.0826	0.5803±0.0652	0.5619±0.0824	0.6377±0.0781	0.6122±0.0221	0.6291±0.0811	0.6408±0.0994
Data-08	0.5241±0.0424	0.6781±0.0523	0.6197±0.0470	0.6775±0.0363	0.7032±0.0591	0.7093±0.0511	0.7214±0.0651	0.7323±0.0524
Data-09	0.5543±0.0242	0.7296±0.0762	0.6510±0.0537	0.7462±0.0562	0.7113±0.0418	0.7791±0.0438	0.7544±0.0448	0.7802±0.0577
Data-10	0.7334±0.078	0.5658±0.1520	0.6141±0.0707	0.7650±0.0446	0.7548±0.0685	0.7569±0.0615	0.7765±0.0545	0.7811±0.0563
Data-11	0.5000±0	0.7847±0.0313	0.7140±0.0411	0.8096±0.0497	0.7931±0.0584	0.7983±0.0764	0.7981±0.0384	0.8121±0.0187
Data-12	0.4998±0.0003	0.5914±0.0821	0.6298±0.0528	0.6639±0.0632	0.6476±0.0711	0.7300±0.0551	0.7018±0.0651	0.7299±0.0336
Data-13	0.5146±0.0283	0.8301±0.0447	0.7888±0.0739	0.8832±0.0368	0.8283±0.0505	0.8766±0.0543	0.8321±0.0485	0.8862±0.0388
Data-14	0.5000±0	0.6982±0.0696	0.7061±0.091	0.7281±0.0702	0.7345±0.0534	0.7297±0.0474	0.7325±0.0424	0.7622±0.0478
Data-15	0.5496±0.0347	0.7616±0.0557	0.4982±0.0562	0.6990±0.0376	0.6954±0.0717	0.7198±0.0657	0.7098±0.0547	0.7301±0.0460
Data-16	0.7726±0.0484	0.9101±0.0243	0.9185±0.0278	0.8887±0.0314	0.8809±0.0471	0.9090±0.0331	0.8909±0.0451	0.9292±0.0288
Data-17	0.5142±0.0455	0.9011±0.0340	0.9372±0.0141	0.9222±0.0335	0.8911±0.0306	0.9112±0.0216	0.9223±0.0546	0.9386±0.0299
Data-18	0.5000±0	0.6081±0.0830	0.5855±0.1113	0.6526±0.1080	0.5343±0.0457	0.6432±0.0387	0.6765±0.0547	0.6859±0.0547
Data-19	0.5445±0.0545	0.6578±0.1120	0.6648±0.0723	0.5407±0.0699	0.6026±0.0828	0.7199±0.0328	0.7232±0.0398	0.7414±0.0150
Data-20	0.5000±0	0.5334±0.1009	0.5702±0.0705	0.6639±0.0915	0.5325±0.0433	0.6329±0.0483	0.6534±0.0723	0.6670±0.0713
Data-21	0.5000±0	0.6628±0.0508	0.6560±0.0654	0.6874±0.0623	0.6421±0.0504	0.6712±0.0344	0.6735±0.0584	0.6884±0.0788
Data-22	0.5000±0	0.6620±0.1065	0.7555±0.0515	0.8012±0.0648	0.7708±0.0781	0.7784±0.0611	0.7654±0.0711	0.8063±0.0694
Data-23	0.5000±0	0.6426±0.0674	0.5962±0.0556	0.6161±0.0508	0.6653±0.0792	0.7124±0.0432	0.6987±0.0892	0.7020±0.0142
Data-24	0.5000±0	0.6584±0.0933	0.6203±0.0806	0.6967±0.0528	0.5653±0.0514	0.6568±0.0484	0.6786±0.0624	0.6689±0.0475
Data-25	0.8974±0.0412	0.9968±0.0025	0.9986±0.0015	0.9982±0.0009	0.7907±0.2334	0.9877±0.0873	0.9876±0.1034	0.9988±0.0010
Data-26	0.4995±0.0004	0.7450±0.1227	0.8308±0.0610	0.8533±0.0917	0.8467±0.0619	0.8678±0.0322	0.8909±0.0547	0.8850±0.0715
Data-27	0.9296±0.0658	0.9637±0.0574	0.9229±0.0494	0.9621±0.0768	0.9633±0.0428	0.9891±0.1088	0.9891±0.0278	0.9795±0.0521
Data-28	0.7571±0.0887	0.9206±0.1127	0.9261±0.0713	0.9647±0.0236	0.8993±0.0497	0.9436±0.0297	0.9513±0.0417	0.9763±0.0440
Data-29	0.5000±0	0.5460±0.0706	0.5251±0.0454	0.5661±0.0361	0.5475±0.0474	0.5329±0.0445	0.5476±0.0454	0.5761±0.0249
Data-30	0.5000±0	0.5516±0.0988	0.5940±0.1184	0.6559±0.1223	0.5722±0.0527	0.6346±0.0327	0.6218±0.0337	0.6681±0.0160
Average	0.5987	0.7287	0.7141	0.7567	0.7290	0.7672	0.7670	0.7854

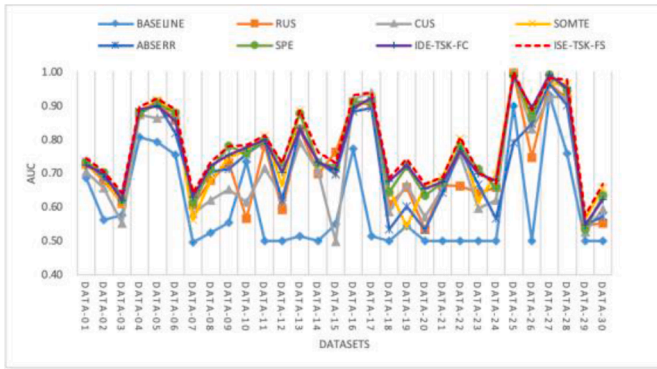
Table 3

Classification performance on 30 real-life datasets in terms of F-score and the standard deviation.

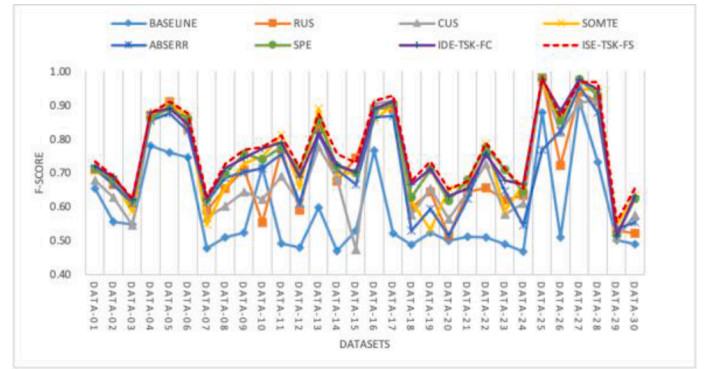
Datasets	0-TSK-FS	RUS	CUS	SMOTE	ABSERR	SPE	IDE-TSK-FC	ISE-TSK-FS
Data-01	0.6543±0.0340	0.7091±0.0222	0.6783±0.0310	0.7054±0.0348	0.713±0.0248	0.7112±0.0542	0.7201±0.0247	0.7341±0.0131
Data-02	0.5547±0.1316	0.6654±0.0388	0.6269±0.0167	0.6756±0.0586	0.6653±0.0379	0.6821±0.0548	0.6881±0.0437	0.6896±0.0231
Data-03	0.5478±0.0524	0.6121±0.0430	0.5456±0.0454	0.5848±0.0400	0.6097±0.0496	0.6198±0.0430	0.6200±0.0444	0.6245±0.0290
Data-04	0.7789±0.0181	0.8689±0.0146	0.8628±0.0228	0.8684±0.0226	0.8542±0.0461	0.8621±0.0436	0.8776±0.0562	0.8765±0.0344
Data-05	0.7589±0.0459	0.9098±0.0326	0.9102±0.0333	0.9075±0.0218	0.8767±0.03	0.8890±0.0562	0.8891±0.0283	0.9091±0.0039
Data-06	0.7452±0.0407	0.8324±0.0646	0.8656±0.0419	0.8658±0.0346	0.8244±0.0726	0.8598±0.0456	0.8434±0.0419	0.8767±0.0293
Data-07	0.4776±0.0994	0.5897±0.0726	0.5673±0.0537	0.5475±0.0824	0.6134±0.0781	0.6224±0.0826	0.6211±0.0652	0.6287±0.0034
Data-08	0.5098±0.0424	0.6543±0.0523	0.6021±0.0470	0.6584±0.0363	0.6854±0.0591	0.6987±0.0523	0.7142±0.0443	0.7267±0.0524
Data-09	0.5231±0.0577	0.7098±0.0762	0.6432±0.0537	0.7276±0.0362	0.7009±0.0418	0.7548±0.0332	0.7432±0.0677	0.7674±0.0142
Data-10	0.7154±0.0563	0.5541±0.152	0.6216±0.0707	0.7486±0.0446	0.7134±0.0685	0.7390±0.1520	0.7711±0.0437	0.7761±0.0107
Data-11	0.4909±0.0187	0.7690±0.0313	0.6896±0.0321	0.8121±0.0497	0.7563±0.0584	0.7768±0.0313	0.7901±0.0411	0.8091±0.0043
Data-12	0.4786±0.0536	0.5891±0.0421	0.6087±0.0528	0.6549±0.0532	0.6987±0.0711	0.7124±0.0451	0.6872±0.0562	0.7167±0.0023
Data-13	0.5986±0.0288	0.8290±0.0447	0.7753±0.0739	0.8897±0.0368	0.8124±0.0505	0.8564±0.0447	0.8187±0.0439	0.8745±0.0238
Data-14	0.4698±0.0578	0.6765±0.0696	0.6834±0.0910	0.6976±0.0702	0.7098±0.0454	0.7198±0.0296	0.7231±0.0610	0.7563±0.0023
Data-15	0.5298±0.0460	0.7436±0.0657	0.4732±0.0526	0.7006±0.0376	0.6653±0.0717	0.6993±0.0557	0.7012±0.0562	0.7289±0.0217
Data-16	0.7654±0.0288	0.8907±0.0243	0.9042±0.0278	0.8549±0.0314	0.8645±0.0471	0.8832±0.0243	0.8876±0.0678	0.9126±0.0324
Data-17	0.5213±0.0299	0.8890±0.0360	0.9126±0.0141	0.8997±0.0235	0.8674±0.0606	0.9009±0.0340	0.9123±0.0141	0.9276±0.0353
Data-18	0.4876±0.0637	0.5983±0.0830	0.5765±0.1530	0.6153±0.1080	0.5298±0.0457	0.6278±0.0830	0.6654±0.0911	0.6765±0.0041
Data-19	0.5234±0.1050	0.6435±0.0620	0.6528±0.0723	0.5321±0.0699	0.5943±0.0828	0.7095±0.1120	0.7112±0.0723	0.7347±0.0345
Data-20	0.4987±0.0663	0.5135±0.0909	0.5643±0.0705	0.6533±0.0915	0.5134±0.0433	0.6176±0.1009	0.6325±0.0705	0.6513±0.0003
Data-21	0.5121±0.0788	0.6435±0.0608	0.6438±0.0764	0.6654±0.0623	0.6246±0.0504	0.6801±0.0508	0.6547±0.0654	0.6764±0.0034
Data-22	0.5098±0.0694	0.6563±0.1065	0.7253±0.0515	0.7865±0.0648	0.7653±0.0781	0.7765±0.1065	0.7547±0.0515	0.7869±0.0021
Data-23	0.4888±0.0842	0.6189±0.0574	0.5769±0.0556	0.5876±0.0508	0.6452±0.0792	0.7091±0.0674	0.6781±0.0556	0.7123±0.0004
Data-24	0.4679±0.0325	0.6341±0.0933	0.6091±0.0646	0.6657±0.0528	0.5452±0.0214	0.6443±0.0933	0.6657±0.0806	0.6543±0.0005
Data-25	0.8787±0.0210	0.9793±0.0045	0.9689±0.0015	0.9675±0.0009	0.7674±0.2334	0.9800±0.0025	0.9768±0.0015	0.9812±0.0112
Data-26	0.5092±0.0455	0.7218±0.1227	0.8190±0.0341	0.8467±0.0917	0.8245±0.0519	0.8546±0.1227	0.8823±0.0610	0.8658±0.0004
Data-27	0.9090±0.0521	0.9376±0.0574	0.9049±0.0494	0.9453±0.0768	0.9534±0.0428	0.9767±0.0574	0.9754±0.0494	0.9672±0.0188
Data-28	0.7321±0.0560	0.9091±0.0927	0.9167±0.0653	0.9453±0.0236	0.8784±0.0227	0.9269±0.1127	0.9456±0.0713	0.9672±0.0387
Data-29	0.5009±0.0449	0.5290±0.0706	0.5086±0.0454	0.5452±0.0361	0.5358±0.0474	0.5176±0.0706	0.5143±0.0454	0.5561±0.0032
Data-30	0.4890±0.0620	0.5218±0.0988	0.5764±0.1184	0.6342±0.1223	0.5532±0.0527	0.6321±0.0988	0.6321±0.1184	0.6567±0.0021
Average	0.5876	0.7134	0.7005	0.7397	0.7091	0.7545	0.7566	0.7741

in scenarios of small data, due to the adopted equal interval division strategy, SPE may generate some “empty blocks”. Thus, under-sampling from all blocks may not balance the minority class. ISE-TSK-FS employs clustering to partition the majority

class into different blocks according to the object sensitivity, which can address the issue of “empty blocks” well. This superiority can be demonstrated from small datasets, such as Data-01, Data-02, Data-03 and so on.

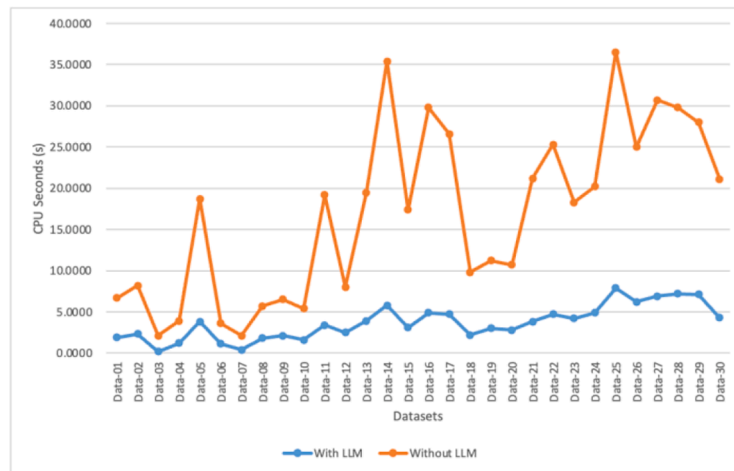


(a) Classification performance in terms of AUC



(b) Classification performance in terms of F-score

. 5. Classification performance on 30 real-world imbalanced datasets.



. 6. CPU seconds w/o LLM.

Table 4
Average ranking values and p-value of Friedman test.

Algorithms	Average ranking values
BASELINE	7.8108
RUS	5.3514
CUS	6.1622
SOMTE	4.2703
ABSERR	5.2162
SPE	3.8243
IDE-TSK-FC	1.8243
ISE-TSK-FS	1.5405
<i>p-value</i> : 0	

too high, the stacked generalization principle may cause a ripple effect due to the incorrect output of the first 0-TSK-FS. The evidences can be found from the results on Data-22 and Data-30.

What's more, Friedman and Holm post-hoc tests [40] are introduced to comprehensively analysis the significant differences between ISE-TSK-FS and the comparison methods. Specifically, Friedman test is first used to detect whether there is a significant difference among all algorithms including ISE-TSK-FS and comparison algorithms over all synthetic and UCI datasets. The results in terms of average ranking values and *p-value* are listed in Table 4. We see that *p-value* computed by Friedman test is 0 which is smaller than 0.05, which suggests that there exist significant different among all algorithms at the 5% significance level.

Next, due to the highest ranking value of ISE-TSK-FS, we fix it as a reference and use Holm test to find out who is significantly different from ISE-TSK-FS from the comparison algorithms. As shown in Table 5, we find that the Holm's procedure rejects all hypotheses which means that all comparison algorithms have significant differences.

4.4. Robustness and sensitivity analysis

(1) Robustness analysis

As we know that the random process during model training will affect the robustness of the model. From Fig. 3, it can be found that RUS in component 1, clustering in component 2 and antecedent learning in 0-

Table 5
Average ranking values and p-value of Friedman test.

<i>i</i>	Algorithms	$Z = (R_0 - R_i)/SE$	<i>p</i>	Holm	Hypothesis
7	BASELINE	11.0102	0	0.007143	Rejected
6	CUS	8.1153	0	0.008333	Rejected
5	RUS	6.6915	0	0.01	Rejected
4	ABSERR	6.4542	0	0.0125	Rejected
3	SOMTE	4.7932	0.000002	0.016667	Rejected
2	SPE	4.0102	0.000061	0.025	Rejected
1	IDE-TSK-FC	.4983	0.618267	0.05	Rejected

(4) IDE-TSK-FC uses the stacked generalization principle as an ensemble strategy. However, for each learning component, i.e., 0-TSK-FS, the input data is imbalanced. If the imbalance ratio is

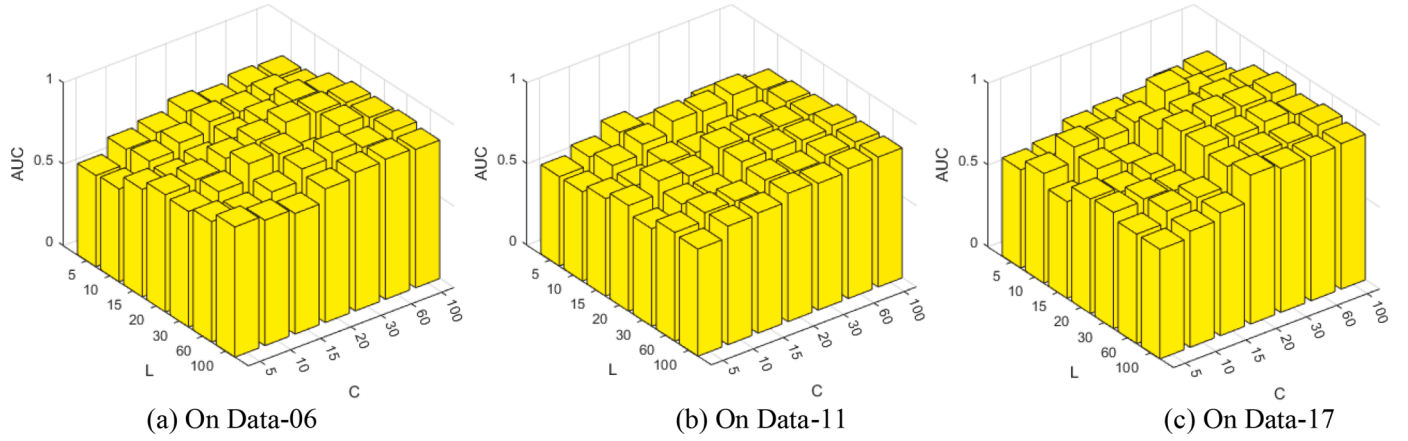


Fig. 7. Sensitivity analysis.

Table 6

Five trained fuzzy rules of one ensemble procedure.

ISE-TSK-FS			
k -th fuzzy rule: If x_{i1} is $\theta_1^k \wedge x_{i2}$ is $\theta_2^k \wedge \dots \wedge x_{id}$ is θ_d^k , then $f^k(x_i) = \beta^k, k = 1, 2, \dots, K$.			
KPCA	Number of fuzzy rules	Antecedent parameters $\mu^k = [\mu_1^k, \mu_2^k, \dots, \mu_d^k]$, $\delta^k = [\delta_1^k, \delta_2^k, \dots, \delta_d^k]$	Consequent parameters $[\beta^k \text{ for class 1 (health)}, \beta^k \text{ for class 2 (epilepsy)}]$
	1	[0.0297, 0.0239, 0.0098, -0.0120, 0.0013, -0.0127] [0.0016, 0.0069, 0.0066, 0.0054, 0.0045, 0.0060]	[0.0126, 0.0642]
	2	[0.0489, 0.0311, -0.0043, -0.0037, 0.0019, -0.0072] [0.0017, 0.0068, 0.0064, 0.0054, 0.0045, 0.0058]	[0.0503, 0.0103]
	3	[0.0108, 0.0689, 0.0455, -0.0037, -0.0345, 0.0912] [0.0017, 0.0093, 0.0074, 0.0063, 0.0061, 0.0090]	[0.0549, 0.0545]
	4	[0.0387, 0.0280, 0.0045, -0.0076, 0.0014, -0.0098] [0.0016, 0.0069, 0.0066, 0.0054, 0.0045, 0.0059]	[0.0258, 0.0293]
	5	[0.0533, 0.0315, -0.0026, -0.0013, 0.0012, -0.0068] [0.0017, 0.0067, 0.0063, 0.0053, 0.0044, 0.0057]	[0.0594, 0.0062]

Algorithm 1

0-TSK-FS.

Input: Training set $D = \{x_i, y_i\}_{i=1}^N$, total number of used fuzzy rules K .
Output: Consequent vector β^T .
Procedure:
 Use FCM to partition D into K clusters, and compute μ_j^k and δ_j^k by

$$\mu_j^k = \frac{\sum_{i=1}^N u_{ik} x_{ij}}{\sum_{i=1}^N u_{ik}}, \text{ and } \delta_j^k = \frac{h \sum_{i=1}^N u_{ik} (x_{ij} - \mu_j^k)^2}{\sum_{i=1}^N u_{ik}},$$
 where u_{ik} represents the fuzzy membership degree of the i th object belonging to the k -th cluster, h is a user-defined parameter (the recommended value is 0.5).
 Construct Ω by (3) as

$$\Omega = \begin{bmatrix} \varphi^1(x_1, \mu^1, \delta^1) & \dots & \varphi^K(x_1, \mu^K, \delta^K) \\ \vdots & & \vdots \\ \varphi^1(x_2, \mu^1, \delta^1) & \dots & \varphi^K(x_2, \mu^K, \delta^K) \end{bmatrix}.$$
 Compute β^T by (6).

TSK-FS may have random processes. This is because RUS is a random under-sampling strategy, and classical K -means and FCM use the random cluster center initialization strategy. In this study, to eliminate randomness and hence improve robustness, we use a non-random cluster center initialization strategy, as shown in Algorithm 3.

Algorithm 2

ISE-TSK-FS.

Input: Training set $D = \{x_i, y_i\}_{i=1}^N$, number of 0-TSK-FS C , number of blocks L .
Output: Final ensemble F .
Procedure:
 01: Initialize U_n as the majority class of D .
 02: Initialize U_p as the minority class of D .
 03: Train an initial 0-TSK-FS $f^{(0)}$ using RUS and set $\beta^{(0)} = 1$.
For $i = 1$ to C
 04: Ensemble $F^{(i)}(x) = \sum_{j=0}^{i-1} \beta^{(j)} f^{(j)}(x) / \sum_{j=0}^{i-1} \beta^{(j)}$.
 05: Compute and normalize sensitivity of all objects in U_n w.r.t $F^{(i)}$ using (8).
 06: Use K -means to cluster U_n into different blocks, $B_l, 1 < l \leq L$ according to the normalized object sensitivity.
 07: Compute average sensitivity of the l -th block as SL_l .
 08: Compute the self-paced factor by $\alpha = \tan(\frac{i\pi}{2C})$.
 09: Unnormalized sampling weight of the l -th block by $w_l = \frac{1}{SL_l + \alpha}$.
 10: Under-sample $\frac{w_l}{\sum_m w_m} |U_p|$ samples from the l -th block.
 11: Train $f^{(i)}$ using U_p and the under-sampled samples from all blocks and record the training error as $\alpha^{(i)}$.
 12: Assign the i th 0-TSK-FS a weight $\beta^{(i)} = 1/\alpha^{(i)}$.
End
 13: **Return** final ensemble $F(x) = \sum_{c=1}^C \beta^{(c)} f^{(c)}(x) / \sum_{c=1}^C \beta^{(c)}$

Algorithm 3

Non-random cluster center initialization strategy.

Input: n input samples x_i , number of clusters C .
Output: Cluster center set V .
Procedure:
 Calculate the first cluster center by $c = \arg \min_{1 \leq j \leq n} \sum_{i=1}^n r(x_j, x_i)$ and $v_1 = x_c$.
 Set $V = \{v_1\}$ and $t = 1$.
 Repeat
 $t = t + 1$.
 Calculate the t -th cluster center by $c = \arg \max_{1 \leq i \leq n, x_i \notin V} \min_{1 \leq c \leq |V|} r(x_c, x_i)$ and $v_t = x_c$.
 $V = V \cup \{v_t\}$.
 Until $(t = C)$

In Algorithm 3, $r(x_j, x_i)$ is a similarity computation function. As we see from Algorithm 3 that the first cluster center is selected from the sample that is most central to the input set. Then the rest cluster centers are selected one by one in such a manner that each cluster center is most dissimilar to all the centers that have been already picked. Algorithm 3 makes the initial cluster centers evenly distributed. What is more, when the input samples are given, the center initialization is always fixed without any randomness.

In the first component of ISE-TSK-FS, although RUS can be replaced

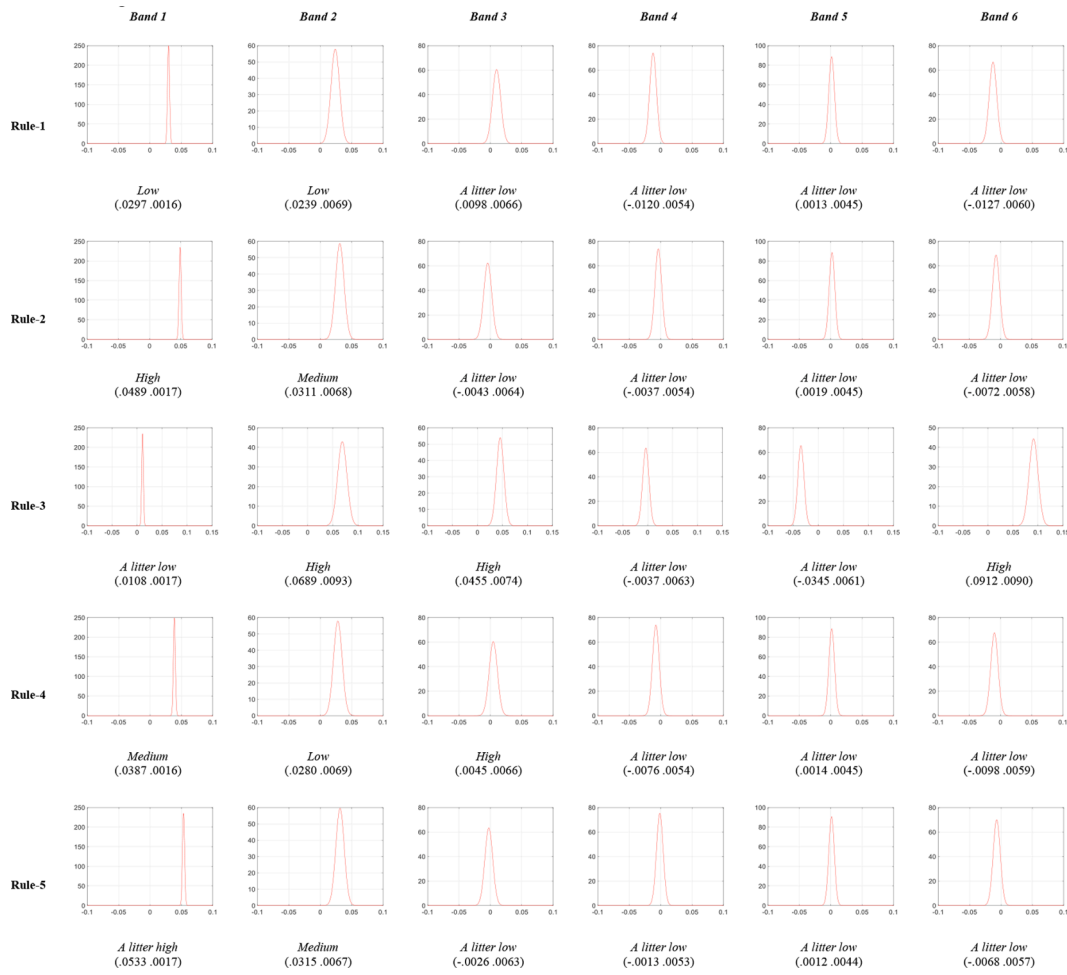


Fig. 8. Fuzzy membership function of each fuzzy rule.

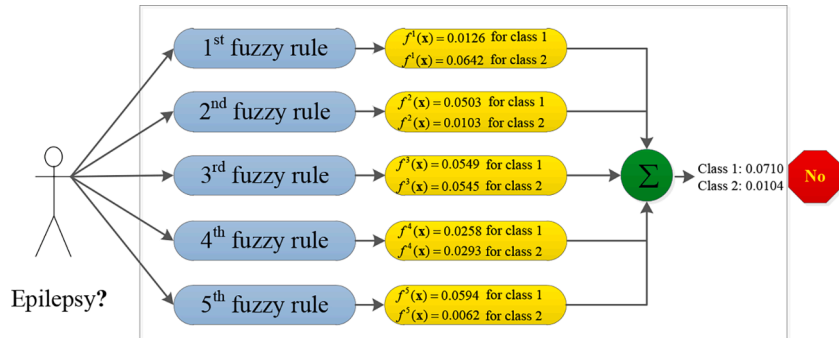


Fig. 9. Interpretable diagnosis procedure.

by other fixed under-sampling strategies, e.g., CUS with the non-random cluster center initialization strategy, we do not do that due to the following three reasons. Firstly, from Algorithm 2, we see that a weighting strategy is used during ensemble, which can offset the effects of inappropriate sampling. Secondly, extra fixed strategy may incur consume more computational resource. Thirdly, RUS in the first component can keep diversity of each learner in ensemble learning, which is a key factor for enhancing the ensemble's performance [41].

(1) Sensitivity analysis

The number of blocks L and the number of 0-TSK-FS C are two key parameters in ISE-TSK-FS. To investigate how these two parameters act on ISE-TSK-FS, we select three datasets, Data-06, Data-11 and Data-17 and observe the performance on them in terms of AUC against L and C . The results are shown in Fig. 7.

It can be found that when C and L exceed 20, the performance of ISE-TSK-FS basically reaches a relatively stable state. That is to say, ISE-TSK-FS is not very sensitive to the two parameters C and L , which makes ISE-TSK-FS have better application ability in actual scenarios.

4.5. Interpretability analysis

As we stated in the introduction part that 0-TSK-FS has high interpretability so that we take it as the basic component of ISE-TSK-FS. To illustrate the interpretability of ISE-TSK-FS, we do a case study of epilepsy diagnosis.

The epilepsy EEG data [42,43] is collected from the University of Bonn, which contains 200 healthy volunteers and 300 epileptic volunteers. We select 50 healthy volunteers and all epileptic volunteers to set up an imbalance binary classification task. We follow the pre-processing steps used in [44] to obtain kernel principle component analysis (KPCA) features for following modeling. After pre-treatment, we finally obtain 6 features in the KPCA feature space, and each feature can be explained as the EEG band energy. The training results (5 trained fuzzy rules) in terms of antecedent and consequent parameters of one assembling procedure are shown in Table 6. Fig. 8 shows the fuzzy membership functions of each rule. According to the center μ^k of each fuzzy membership function, we introduce five semantics, they are “a little low”, “low”, “medium”, “a little high” and “high”, respectively. Thus, as shown in Fig. 5, each fuzzy membership function can be endowed with certain semantics. For instance, the fuzzy mapping of the first band in the first fuzzy rule can be explained as “the energy of an EEG signal band is “low”. Notably, this semantic description is subjective because it depends on the clinical knowledge domain experts have. For instance, the first trained fuzzy rule can be described as:

If band 1 is “low”, and band 2 is “low”, band 3 is “a little low”, and band 4 is “a little low”, and band 5 is “a little low”, and band 6 is “a little low”,

Then the consequent of the first fuzzy rule can be expressed as:

$$f^1(\mathbf{x}) = 0.0126 \text{ for class 1 and } f^1(\mathbf{x}) = 0.0642 \text{ for class 2.}$$

With the trained 5 fuzzy rules, for a patient, whether he/she has epilepsy can be determined by the procedure shown in Fig. 9.

5. Conclusion

TSK fuzzy systems are renowned for striking an excellent compromise between strong interpretability and great nonlinear approximation performance. So far, they have been extensively used in many scenarios. However, multiple studies [13, 26,27,29–31] have highlighted their poor performance on imbalanced data where the class distribution is substantially unequal. Under this circumstance, TSK fuzzy systems are inclined to be biased toward the majority class, leading to a poor classification rate for the minority class, which in fact are often more of our concern and interest. In this study, a new TSK fuzzy system ISE-TSK-FS at the level of a sensitivity ensemble is proposed. To extensively evaluate the ISE-TSK-FS's performance on different types of imbalanced datasets, 7 synthetic datasets and 30 real-life datasets from the UCI repository were employed in the experiments. Additionally, a case study of epilepsy recognition is performed to exhibit the interpretability of ISE-TSK-FS.

Our study still has some limitations. The basic component 0-TSK-FS is actually a single-layer feedforward neural network (SLFNN), see Fig. 1. Thus, although it has high interpretability, it inevitably inherits some shortcomings from SLFNN. For instance, it is difficult to wisely determine the optimal network structure. Additionally, in our ensemble framework, we record the training error as the weight of each classifier. Although the weighting is straightforward, it is a local strategy. Therefore, to improve ISE-TSK-FS, there are at least two things we can do in the future: 1) in light of the evidence shown in recent studies [45, 46] that the dropout strategies used in deep learning can ensure an optimal network structure, similar strategies are anticipated to lighten the burden of finding the optimal number of fuzzy rules. 2) Like the weighting strategy used in [47], similar new global weighting strategies that weight learning is taken as optimization objective are expected.

CRedit authorship contribution statement

Yuanpeng Zhang: Methodology, Writing – original draft. **Guanjin Wang:** Investigation. **Xiuyu Huang:** Data curation. **Weiping Ding:** Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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